# Heatmaps for Speedup

### Cores: 2

| Width (Qubits) \ Depth (Gates) | 10000 | 20000 | 30000 | 40000 | 50000 | 60000 | 70000 | 80000 | 90000 | 100000 |
|---|---|---|---|---|---|---|---|---|---|---|
| 200 | 1.5 | 1.7 | 1.8 | 1.8 | 1.8 | 1.8 | 1.9 | 1.9 | 1.9 | 1.9 |
| 180 | 1.6 | 1.8 | 1.9 | 1.9 | 2.0 | 1.9 | 1.9 | 1.9 | 1.8 | 1.9 |
| 160 | 1.7 | 1.8 | 1.9 | 1.9 | 1.9 | 1.8 | 1.9 | 1.8 | 1.8 | 2.0 |
| 140 | 1.8 | 1.8 | 1.9 | 2.0 | 2.0 | 2.0 | 1.9 | 2.0 | 2.0 | 2.0 |
| 120 | 1.8 | 1.9 | 1.9 | 2.0 | 1.8 | 2.0 | 1.9 | 1.8 | 1.9 | 2.1 |
| 100 | 2.0 | 1.9 | 1.9 | 1.9 | 1.9 | 1.9 | 1.9 | 1.9 | 2.0 | 2.0 |
| 80 | 1.9 | 2.0 | 2.0 | 2.0 | 1.9 | 1.9 | 1.9 | 1.9 | 2.0 | 1.9 |
| 60 | 1.9 | 2.1 | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 |
| 40 | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 | 1.9 | 1.7 | 2.0 |
| 20 | 1.7 | 1.9 | 1.8 | 2.1 | 2.0 | 2.1 | 1.9 | 2.0 | 2.0 | 1.8 |

### Cores: 4

| Width \ Depth (Gates) | 10000 | 20000 | 30000 | 40000 | 50000 | 60000 | 70000 | 80000 | 90000 | 100000 |
|---|---|---|---|---|---|---|---|---|---|---|
| 200 | 2.3 | 2.8 | 3.1 | 3.2 | 3.3 | 3.1 | 3.6 | 3.4 | 2.9 | 3.4 |
| 180 | 2.6 | 3.0 | 3.3 | 3.3 | 3.5 | 3.3 | 3.4 | 3.5 | 3.7 | 3.6 |
| 160 | 2.8 | 3.2 | 2.6 | 3.5 | 3.6 | 3.3 | 3.6 | 3.7 | 4.0 | 3.8 |
| 140 | 3.0 | 3.3 | 3.4 | 3.6 | 3.5 | 3.7 | 3.7 | 3.6 | 3.3 | 3.6 |
| 120 | 2.6 | 3.6 | 3.5 | 3.5 | 3.6 | 3.7 | 3.7 | 3.8 | 3.6 | 3.6 |
| 100 | 3.2 | 3.7 | 3.8 | 3.7 | 3.7 | 3.7 | 3.7 | 3.9 | 3.9 | 3.7 |
| 80 | 3.3 | 3.5 | 3.7 | 3.9 | 3.7 | 3.5 | 2.7 | 3.1 | 3.6 | 3.9 |
| 60 | 3.5 | 3.8 | 3.9 | 3.5 | 3.5 | 3.4 | 3.5 | 3.5 | 3.7 | 3.7 |
| 40 | 3.1 | 3.5 | 3.7 | 3.5 | 3.6 | 3.7 | 3.3 | 3.7 | 3.8 | 3.9 |
| 20 | 2.6 | 2.8 | 3.3 | 3.6 | 3.4 | 3.8 | 3.6 | 3.6 | 3.7 | 3.5 |

### Cores: 6

| Width \ Depth (Gates) | 10000 | 20000 | 30000 | 40000 | 50000 | 60000 | 70000 | 80000 | 90000 | 100000 |
|---|---|---|---|---|---|---|---|---|---|---|
| 200 | 3.0 | 3.7 | 4.3 | 4.3 | 4.7 | 4.6 | 4.2 | 4.3 | 4.8 | 3.9 |
| 180 | 2.3 | 4.0 | 4.3 | 4.5 | 4.7 | 4.8 | 4.8 | 5.0 | 5.1 | 4.9 |
| 160 | 3.6 | 4.3 | 4.7 | 4.5 | 4.8 | 5.2 | 4.9 | 3.2 | 5.0 | 5.1 |
| 140 | 4.0 | 4.5 | 4.9 | 4.5 | 4.6 | 4.6 | 4.9 | 5.2 | 5.3 | 5.4 |
| 120 | 4.2 | 4.6 | 5.1 | 5.0 | 5.1 | 5.2 | 4.7 | 5.3 | 5.2 | 5.4 |
| 100 | 4.6 | 4.9 | 5.2 | 5.3 | 5.5 | 5.1 | 4.9 | 5.0 | 4.9 | 5.0 |
| 80 | 4.4 | 5.0 | 5.4 | 5.1 | 5.5 | 5.2 | 5.4 | 5.5 | 5.6 | 5.9 |
| 60 | 4.4 | 5.3 | 5.3 | 5.5 | 5.5 | 5.5 | 5.5 | 5.5 | 5.6 | 5.4 |
| 40 | 4.5 | 4.9 | 5.3 | 5.3 | 5.2 | 4.6 | 5.4 | 5.6 | 5.7 | 5.5 |
| 20 | 3.7 | 4.9 | 5.2 | 4.7 | 5.2 | 5.3 | 5.2 | 5.2 | 5.6 | 5.5 |

### Cores: 8

| Width (Qubits) \ Depth (Gates) | 10000 | 20000 | 30000 | 40000 | 50000 | 60000 | 70000 | 80000 | 90000 | 100000 |
|---|---|---|---|---|---|---|---|---|---|---|
| 200 | 3.2 | 4.0 | 4.8 | 5.1 | 5.9 | 6.4 | 5.8 | 5.9 | 5.9 | 5.9 |
| 180 | 3.6 | 4.8 | 5.3 | 5.7 | 5.9 | 6.4 | 5.8 | 6.1 | 6.2 | 6.4 |
| 160 | 3.9 | 5.1 | 6.1 | 5.3 | 5.6 | 5.8 | 6.3 | 5.8 | 6.3 | 6.4 |
| 140 | 4.8 | 4.9 | 6.2 | 6.2 | 5.4 | 6.6 | 6.8 | 6.9 | 5.1 | 7.2 |
| 120 | 4.2 | 4.9 | 5.7 | 6.1 | 6.4 | 6.5 | 6.5 | 7.0 | 6.3 | 6.9 |
| 100 | 5.8 | 5.8 | 6.7 | 6.4 | 6.9 | 7.0 | 6.6 | 6.5 | 7.2 | 6.4 |
| 80 | 5.8 | 7.0 | 6.2 | 6.4 | 7.0 | 7.3 | 6.8 | 6.8 | 6.9 | 7.1 |
| 60 | 6.2 | 6.5 | 6.6 | 7.1 | 6.9 | 6.5 | 6.1 | 6.4 | 6.3 | 6.5 |
| 40 | 3.2 | 5.5 | 6.9 | 5.3 | 7.2 | 7.1 | 6.8 | 7.1 | 7.4 | 7.3 |
| 20 | 2.9 | 3.8 | 6.0 | 6.1 | 6.1 | 6.0 | 6.0 | 6.1 | 7.0 | 6.6 |

### Cores: 10

| Width \ Depth (Gates) | 10000 | 20000 | 30000 | 40000 | 50000 | 60000 | 70000 | 80000 | 90000 | 100000 |
|---|---|---|---|---|---|---|---|---|---|---|
| 200 | 3.4 | 4.6 | 5.6 | 5.9 | 6.7 | 6.5 | 7.1 | 5.8 | 7.1 | 7.5 |
| 180 | 4.1 | 5.5 | 6.2 | 6.1 | 6.2 | 6.6 | 7.1 | 7.4 | 7.1 | 6.8 |
| 160 | 4.7 | 5.7 | 5.8 | 7.2 | 5.8 | 7.5 | 6.1 | 7.7 | 7.4 | 8.3 |
| 140 | 4.3 | 6.2 | 6.0 | 7.2 | 7.1 | 7.6 | 8.2 | 8.2 | 8.2 | 8.6 |
| 120 | 5.9 | 6.8 | 7.5 | 7.5 | 7.2 | 6.7 | 7.8 | 7.9 | 8.1 | 8.3 |
| 100 | 6.9 | 6.5 | 8.2 | 8.3 | 7.9 | 7.4 | 7.9 | 8.0 | 8.4 | 8.5 |
| 80 | 6.4 | 7.7 | 8.1 | 8.0 | 8.0 | 8.5 | 7.8 | 8.1 | 7.7 | 8.1 |
| 60 | 5.1 | 6.4 | 7.6 | 8.5 | 7.4 | 8.0 | 7.7 | 9.1 | 8.1 | 9.0 |
| 40 | 7.0 | 7.0 | 7.0 | 8.2 | 8.3 | 9.0 | 8.2 | 9.5 | 7.9 | 8.0 |
| 20 | 4.1 | 6.2 | 4.3 | 7.7 | 7.6 | 7.6 | 7.1 | 8.6 | 7.7 | 7.8 |

### Cores: 12

| Width \ Depth (Gates) | 10000 | 20000 | 30000 | 40000 | 50000 | 60000 | 70000 | 80000 | 90000 | 100000 |
|---|---|---|---|---|---|---|---|---|---|---|
| 200 | 3.2 | 5.1 | 6.4 | 7.1 | 7.5 | 7.2 | 7.4 | 7.6 | 7.8 | 7.2 |
| 180 | 4.5 | 6.2 | 6.6 | 7.5 | 7.1 | 8.7 | 8.0 | 8.1 | 8.1 | 8.0 |
| 160 | 5.1 | 6.4 | 7.4 | 7.0 | 7.6 | 8.7 | 7.5 | 8.8 | 9.4 | 9.0 |
| 140 | 5.5 | 6.9 | 8.1 | 7.9 | 8.7 | 8.8 | 7.1 | 10.0 | 9.6 | 9.4 |
| 120 | 6.6 | 8.2 | 7.7 | 8.4 | 8.7 | 8.5 | 8.8 | 8.7 | 10.4 | 8.5 |
| 100 | 6.8 | 7.9 | 8.6 | 9.1 | 9.2 | 9.3 | 9.5 | 9.9 | 9.7 | 9.8 |
| 80 | 6.3 | 9.0 | 8.9 | 9.9 | 9.7 | 9.5 | 10.2 | 9.4 | 10.1 | 10.0 |
| 60 | 7.0 | 8.5 | 8.5 | 8.8 | 9.1 | 9.4 | 8.8 | 9.2 | 9.4 | 9.5 |
| 40 | 7.0 | 8.4 | 8.1 | 8.6 | 9.0 | 8.6 | 9.0 | 9.1 | 9.3 | 10.1 |
| 20 | 5.8 | 6.9 | 7.8 | 8.5 | 7.9 | 8.3 | 8.6 | 9.2 | 8.9 | 9.5 |

### Cores: 14

| Width (Qubits) \ Depth (Gates) | 10000 | 20000 | 30000 | 40000 | 50000 | 60000 | 70000 | 80000 | 90000 | 100000 |
|---|---|---|---|---|---|---|---|---|---|---|
| 200 | 3.6 | 5.3 | 6.5 | 6.6 | 8.1 | 7.3 | 7.9 | 7.3 | 8.7 | 9.5 |
| 180 | 3.5 | 6.5 | 7.6 | 8.0 | 9.0 | 8.8 | 9.6 | 9.2 | 8.7 | 9.8 |
| 160 | 5.5 | 7.6 | 8.4 | 8.2 | 9.3 | 9.7 | 8.8 | 9.8 | 10.0 | 10.6 |
| 140 | 5.8 | 7.4 | 7.0 | 9.3 | 9.2 | 10.1 | 10.3 | 10.4 | 10.0 | 9.9 |
| 120 | 7.3 | 9.4 | 9.2 | 9.0 | 10.0 | 8.3 | 10.6 | 11.1 | 11.7 | 10.5 |
| 100 | 8.5 | 8.9 | 9.2 | 10.6 | 9.8 | 9.1 | 10.5 | 10.9 | 11.2 | 11.0 |
| 80 | 8.5 | 9.7 | 8.3 | 11.2 | 11.1 | 10.4 | 11.0 | 10.0 | 11.1 | 11.1 |
| 60 | 8.4 | 9.7 | 10.6 | 10.6 | 10.2 | 9.2 | 11.3 | 11.0 | 11.3 | 10.5 |
| 40 | 8.3 | 7.8 | 10.3 | 10.9 | 10.4 | 8.8 | 10.6 | 10.9 | 11.1 | 9.5 |
| 20 | 6.2 | 8.5 | 9.4 | 9.0 | 10.1 | 9.5 | 9.4 | 11.4 | 11.3 | 9.8 |

### Cores: 16

| Width \ Depth (Gates) | 10000 | 20000 | 30000 | 40000 | 50000 | 60000 | 70000 | 80000 | 90000 | 100000 |
|---|---|---|---|---|---|---|---|---|---|---|
| 200 | 4.0 | 5.5 | 6.5 | 8.1 | 8.6 | 8.7 | 9.1 | 7.5 | 8.1 | 10.4 |
| 180 | 4.8 | 6.9 | 7.6 | 8.7 | 9.3 | 9.6 | 9.9 | 10.6 | 9.2 | 8.6 |
| 160 | 5.6 | 8.0 | 8.7 | 9.4 | 10.5 | 10.3 | 9.0 | 12.4 | 10.5 | 11.2 |
| 140 | 5.6 | 8.5 | 10.0 | 9.4 | 9.9 | 11.0 | 9.6 | 10.5 | 12.7 | 12.1 |
| 120 | 7.6 | 9.3 | 10.8 | 11.3 | 10.9 | 11.5 | 9.6 | 11.3 | 12.4 | 11.7 |
| 100 | 8.1 | 10.5 | 10.3 | 9.9 | 12.1 | 10.1 | 11.0 | 12.4 | 12.6 | 12.9 |
| 80 | 8.0 | 10.0 | 11.1 | 10.3 | 12.6 | 12.5 | 11.9 | 11.5 | 12.4 | 12.0 |
| 60 | 8.7 | 10.0 | 11.2 | 12.4 | 11.8 | 11.4 | 12.1 | 12.6 | 12.1 | 12.6 |
| 40 | 8.3 | 10.3 | 11.4 | 11.4 | 12.0 | 10.4 | 10.8 | 11.6 | 11.7 | 10.3 |
| 20 | 5.5 | 8.3 | 8.7 | 10.5 | 11.5 | 9.8 | 9.1 | 11.3 | 12.2 | 11.9 |

Depth (Gates)

Speedup